

SOSVA, Russian Federation

WMO: 236250

Lat: 63.650N

Lon: 62.100E

Elev: 28

StdP: 100.99

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-41.4	-38.6	-44.0	0.0	-41.3	-41.1	0.1	-38.3	4.9	-8.5	4.3	-9.7	0.2	120	0.635

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	28.5	19.6	26.5	18.3	24.4	17.3	21.0	26.7	19.6	24.5	18.3	22.9	1.4	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.0	13.8	23.6	17.7	12.7	22.4	16.3	11.7	21.2	61.0	26.7	56.2	24.5	51.9	22.9	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.2	4.4	3.7	DB	-41.8	31.2	4.1	1.6	-44.7	32.4	-47.1	33.3	-49.5	34.2	-52.5	35.3	(4)
(5)			WB	-41.7	22.6	3.8	1.6	-44.4	23.8	-46.6	24.7	-48.7	25.6	-51.5	26.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.1	-21.9	-19.3	-9.3	-1.4	6.1	13.4	17.2	13.4	7.6	-0.1	-12.8	-18.7	(6)
(7)		DBStd	15.06	8.99	8.89	7.06	5.89	5.39	5.02	4.29	3.99	4.04	5.40	9.76	10.28	(7)
(8)		HDD10.0	4915	988	821	598	342	148	22	2	12	94	314	683	891	(8)
(9)		HDD18.3	7502	1247	1054	856	591	379	162	74	162	322	572	933	1149	(9)
(10)		CDD10.0	517	0	0	0	0	28	124	226	116	22	1	0	0	(10)
(11)		CDD18.3	61	0	0	0	0	1	13	39	8	0	0	0	0	(11)
(12)		CDH23.3	657	0	0	0	0	27	141	416	72	2	0	0	0	(12)
(13)		CDH26.7	154	0	0	0	0	3	28	112	12	0	0	0	0	(13)
(14)	Wind	WSAvg	1.3	1.0	1.1	1.5	1.6	1.7	1.5	1.2	1.4	1.4	1.1	1.1	(14)	
(15)	Precipitation	PrecAvg	467	26	20	24	29	38	64	63	52	38	34	24	(15)	
(16)		PrecMax	638	50	38	61	71	117	123	158	138	113	85	78	42	(16)
(17)		PrecMin	284	5	1	1	1	4	20	8	18	17	9	1	10	(17)
(18)		PrecStd	93	13	10	15	14	26	30	43	31	25	17	17	7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.9	1.4	7.4	14.6	29.7	31.7	28.2	21.9	14.4	3.9	1.7	(19)	
(20)			MCWB	-2.6	-0.1	2.8	8.3	14.8	18.8	21.5	20.2	16.3	10.7	2.3	0.2	(20)
(21)		2%	DB	-3.6	-1.2	4.4	11.3	23.0	26.9	29.7	25.2	18.3	11.3	2.7	0.1	(21)
(22)			MCWB	-4.2	-2.4	1.2	5.8	13.5	17.6	20.3	18.8	13.9	8.4	1.4	-1.4	(22)
(23)		5%	DB	-5.5	-4.0	2.5	8.7	19.3	24.7	27.8	22.7	16.0	8.9	1.2	-2.2	(23)
(24)			MCWB	-6.2	-5.2	0.0	4.2	11.7	16.7	19.4	17.4	12.5	6.9	0.0	-3.2	(24)
(25)		10%	DB	-8.4	-6.1	0.7	6.8	15.7	22.3	25.7	20.4	14.1	6.6	-0.5	-4.6	(25)
(26)			MCWB	-9.1	-7.1	-1.4	3.1	9.6	15.6	18.2	16.0	11.3	5.1	-1.5	-5.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.4	0.0	3.2	8.6	16.6	20.5	23.1	21.9	18.3	11.2	2.7	0.5	(27)
(28)			MCDWB	-2.1	1.3	6.3	13.8	22.9	26.1	28.6	26.9	20.1	13.9	3.5	1.5	(28)
(29)		2%	WB	-4.2	-2.3	1.5	6.1	13.9	19.0	21.7	19.8	14.4	8.7	1.5	-1.4	(29)
(30)			MCDWB	-3.8	-1.2	4.0	10.6	20.8	24.8	27.8	23.7	17.7	10.9	2.5	0.0	(30)
(31)		5%	WB	-6.2	-5.1	0.1	4.7	11.9	17.7	20.5	18.1	12.7	6.9	0.1	-3.2	(31)
(32)			MCDWB	-5.5	-4.2	2.3	7.9	18.8	23.2	26.1	21.6	15.3	8.8	1.2	-2.3	(32)
(33)		10%	WB	-9.0	-7.1	-1.2	3.5	10.0	16.2	19.2	16.5	11.5	5.2	-1.6	-5.3	(33)
(34)			MCDWB	-8.3	-6.0	0.6	6.3	15.2	21.3	23.9	20.0	13.8	6.5	-0.5	-4.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.2	9.6	11.0	10.0	9.9	9.7	10.7	9.6	7.7	6.3	7.0	8.1	(35)
(36)			MCDBR	7.8	9.3	9.9	12.0	15.9	13.0	13.7	12.8	10.6	8.2	6.2	8.5	(36)
(37)			MCWBR	7.4	8.3	7.5	7.9	8.2	6.0	6.1	7.1	6.6	6.0	5.5	7.7	(37)
(38)		5% WB	MCDBR	7.7	8.6	9.7	10.2	14.6	11.8	11.8	11.0	9.8	8.0	5.9	8.3	(38)
(39)			MCWBR	7.4	8.0	7.6	7.0	7.9	6.4	6.3	7.0	6.6	6.1	5.4	7.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.226	0.283	0.324	0.356	0.362	0.374	0.417	0.386	0.339	0.302	0.226	0.225	(40)	
(41)		taud	1.934	2.045	2.049	2.110	2.361	2.417	2.284	2.354	2.455	2.345	1.992	1.914	(41)	
(42)		Ebn at Noon	343	613	742	810	850	846	788	781	753	618	339	132	(42)	
(43)		Edh at Noon	31	69	104	122	105	102	114	97	72	52	28	15	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.20	0.94	2.42	4.02	5.08	5.28	5.35	3.67	2.02	0.92	0.26	0.08	(44)	
(45)		RadStd	0.01	0.07	0.17	0.44	0.44	0.42	0.41	0.46	0.38	0.27	0.10	0.03	0.00	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page